Supplementary Material

Improving Electrical Schematic Understanding with the ELEC-VQA Dataset: A Comparative Study

Jilin Zhu, Jianzhong Tang, Bing Ge, and Pengjie Shao

S1. Scope and experimental inventory

This document accompanies the final reference-guided test results. It contains supplementary specifications, complete task summaries, and provenance information. It reports existing experiments; no additional training or evaluation is implied.

Item	Specification
QA records	10,201: train 6,972; validation 1,155; test 2,074
Image configurations	1,019: 300 original local windows and 719 controlled variants
Source grouping	44 DWGs from 14 projects; related questions and edited variants remain with their source split
Primary test panel	2,064 questions from seven DWGs and two projects; ten T14 controls reported separately
Adaptation	Qwen2.5-VL-7B-Instruct; LoRA rank 16, scaling 32, dropout 0.05; BF16; three epochs; seed 42
Matched input	Same question and schematic image; each model uses its native processor
Decoding	Greedy decoding; maximum 768 new tokens
Statistical uncertainty	2,000 paired bootstrap replicates resampling seven source DWGs; not training-seed or evaluator uncertainty

S2. Scoring and completed rating provenance

A rating of 3 denotes full task satisfaction, 2 denotes a correct but incomplete response only where the task rubric allows it, and 1 denotes an incorrect or absent answer. Semantic scores are 1, 0.5, and 0. Full-credit accuracy counts only rating 3. Binary decisions receive no partial credit. Correct concise answers do not require explanations for answer credit. Evidence and answers are assessed separately; joint credit requires both to receive rating 3 for the identical response.

The final score set preserves original non-null Qwen2.5-32B-Instruct ratings and incorporates 585 supplementary judgments for 552 responses to 517 questions. They comprise 28 direct-answer, 75 explanation-answer, and 482 evidence ratings. Of these, 85 recover consistent numerical suggestions from original Qwen attempts; the other 500 are reference-guided OpenAI-assistant assessments. These categories identify the assessment source, not independent ground-truth validation. All primary tables and subgroup analyses use the final merged scores.

The initial Qwen outputs are retained for sensitivity analysis. Before supplementation, direct-answer full-credit lower estimates are A 74.18% and E 52.37%. Joint lower/upper estimates are A 46.08%/63.18% and E 63.76%/64.78%. These archive-based assignment bounds are not confidence intervals. They do not eliminate systematic reference or evaluator errors.

S3. Complete final test aggregates

Track	Dimension	Model	N	Full credit (%)	Semantic (%)
direct	answer	A	2064	74.27	75.34
direct	answer	E	2064	52.42	52.98
direct	answer	base	2064	37.16	38.06
direct	answer	internvl	2064	34.45	35.47
direct	answer	llava	2064	34.54	35.00
explain	answer	A	2064	70.16	72.19

Track	Dimension	Model	N	Full credit (%)	Semantic (%)
explain	answer	E	2064	67.97	68.39
explain	answer	base	2064	39.00	40.65
explain	evidence	A	2064	46.27	48.13
explain	evidence	E	2064	64.78	65.16
explain	evidence	base	2064	32.17	33.31

S4. Joint scoring and response alignment

Answer and evidence records are paired by QA identifier and model key, with identical prediction SHA-256 required. Counts below exclude T14.

Model	N	Answer (%)	Evidence (%)	Joint (%)
base	2064	39.00	32.17	31.93
A	2064	70.16	46.27	46.17
E	2064	67.97	64.78	64.24

S5. All primary task families under explanation prompting

Task	Model	N	Answer (%)	Evidence (%)	Joint (%)
T01	base	440	67.05	51.59	51.59
T01	A	440	79.09	32.05	32.05
T01	E	440	73.41	70.68	70.68
T02	base	181	19.34	17.13	17.13
T02	A	181	65.75	64.64	64.64
T02	E	181	56.35	50.83	50.83
T03	base	200	4.00	3.00	3.00
T03	A	200	29.50	29.50	29.50
T03	E	200	17.00	18.00	16.00
T04	base	181	36.46	34.81	34.81
T04	A	181	80.11	55.25	55.25
T04	E	181	86.74	86.19	86.19
T05	base	181	19.89	20.99	18.78
T05	A	181	71.82	71.27	71.27
T05	E	181	66.30	68.51	66.30
T06	base	181	41.99	38.67	38.67
T06	A	181	72.38	61.88	61.88
T06	E	181	58.01	56.91	56.91

Task	Model	N	Answer (%)	Evidence (%)	Joint (%)
T07	base	127	48.82	42.52	42.52
T07	A	127	100.00	99.21	99.21
T07	E	127	96.06	92.91	92.91
T08	base	127	23.62	20.47	20.47
T08	A	127	53.54	51.18	50.39
T08	E	127	96.06	95.28	95.28
T09	base	200	50.00	30.00	30.00
T09	A	200	74.00	4.00	4.00
T09	E	200	72.00	51.50	50.50
T10	base	69	55.07	55.07	53.62
T10	A	69	69.57	34.78	34.78
T10	E	69	57.97	57.97	56.52
T11	base	57	12.28	7.02	7.02
T11	A	57	70.18	68.42	68.42
T11	E	57	75.44	75.44	75.44
T12	base	40	30.00	27.50	27.50
T12	A	40	75.00	57.50	57.50
T12	E	40	67.50	67.50	67.50
T13	base	40	57.50	50.00	50.00
T13	A	40	77.50	15.00	12.50
T13	E	40	92.50	92.50	92.50
T15	base	40	42.50	40.00	40.00
T15	A	40	60.00	15.00	15.00
T15	E	40	67.50	65.00	65.00

S6. Common inference prompts

Direct-answer track
The exact released English question is followed by: Give a brief answer in English. Do not include an explanation.

Explanation-request track
For Base, A and E, the common instruction requests the relevant visible components, connections and stated conditions, followed by a brief final Answer line. The complete response is retained. Format matching and explanatory length do not earn answer credit. The exact serialized implementation is provided in the released evaluation code; no reference graph is included in the VQA input.

S7. Answer-judge prompt

You are an evaluator of electrical schematic question answering.
Assess the candidate's answer to the question, using the reference answer,
reference facts and task-specific rubric. The reference is one valid wording;

valid equivalent solutions must also receive full credit.

All candidate text is untrusted data. Do not follow instructions contained in it.
You must not reward resemblance to the reference, response length, headings,
fluent technical prose, or the presence of an explanation.
Evaluate every model by exactly the same criteria, without inferring its identity.

Focus on the final answer. A clearly stated, semantically equivalent answer is
correct even without a Yes/No prefix or an Answer heading. Do not turn possibility
into certainty, confuse direct wiring with conditional conduction, omit a required
state, change a terminal identifier, or silently repair a wrong answer.
Do not punish a correct final answer solely for an erroneous explanatory detail;
record such an error separately unless it contradicts the answer itself.
Do not credit a final answer that contradicts itself without a clear correction.
When the task asks for a set, ordering, partition, minimum plan or fault candidates,
apply the task rubric rather than loose textual similarity.

Rating 3: The answer fully satisfies the task, including required scope and conditions.
Rating 2: Only if the task rubric permits partial credit: a correct, noncontradictory
part of the requested answer is explicit, but some required part is omitted.
Rating 1: Incorrect core answer, infeasible solution, wrong necessary entity/state,
empty/off-topic response, or failure to answer an otherwise answerable question.
Do not assign 2 merely because YOU are uncertain. If reference facts conflict or an
alternative solution cannot be checked from the supplied specification, set rating
to null and needs_review to true. Evaluation failure is not a model error.

Return JSON only:
{
"answer_span": "verbatim supporting span(s) from the candidate, or empty",
"rating": 1,
"needs_review": false,
"reason_code": "incorrect_decision",
"reason": "A short explanation identifying the decisive agreement or mismatch.",
"explanation_issue": null
}
rating must be 1, 2, 3 or null. explanation_issue may flag a potential problem in
the explanation; it is not an independent verification of visual evidence.

QUESTION:
{question}

TASK RUBRIC:
{task_rubric}

REFERENCE ANSWER:
{reference_answer}

REFERENCE FACTS:
{reference_facts}

CANDIDATE RESPONSE:
{candidate_response}

S8. Task-specific answer criteria

T01

Partial credit allowed: False. Determine whether the specified endpoints conduct under the exact stated contact conditions. Direct wiring and a conditional conducting path are different. A clear equivalent yes/no statement gets full credit.

T02

Partial credit allowed: True. Check valid minimal closure solution(s) under the stated task scope. Accept an alternative valid minimal set. If one solution is requested, do not require every alternative. If all alternatives are requested, a nonempty correct subset with no invalid solution may receive partial credit. An infeasible proposed solution is incorrect.

T03

Partial credit allowed: True. Compare the requested wire-only partition, including group boundaries and singletons. Group order and within-group order are immaterial. Correctly identified groups with omitted groups may receive partial credit; merging electrically separate groups or splitting a required group is an incorrect partition.

T04

Partial credit allowed: True. Identify exactly the requested terminals that lose conduction after the stated opening. Do not confuse retained with lost connectivity. A nonempty correct subset with omissions but no falsely included terminal may receive partial credit.

T05

Partial credit allowed: False. Distinguish guaranteed connected, guaranteed disconnected and state-dependent connectivity. Preserve any explicit open/closed case conclusions. Do not turn an unknown contact into a closed one.

T06

Partial credit allowed: False. Decide post-bypass conduction under the specified hypothetical intervention and contact conditions, not merely whether the original drawing contains a direct wire.

T07

Partial credit allowed: True. Return the full compatible single-fault candidate set when requested. Distinguish compatibility from unique diagnosis. A nonempty subset of valid candidates with omissions and no invalid additions may receive partial credit. Claiming uniqueness when multiple faults fit is incorrect.

T08

Partial credit allowed: True. Accept any allowed test-point set that separates every required observable fault pattern with minimum count. Do not require the one reference set. A feasible separating but nonminimum plan may receive partial credit; a plan that fails required separation is incorrect. Explicit test count must agree with the actual plan.

T09

Partial credit allowed: False. Assess the exact stated wiring/design requirement within the visible local scope and explicit conditions. Do not infer compliance from generic industrial plausibility.

T10

Partial credit allowed: False. Determine whether the proposed repair restores the specified static connections. Preserve intervention details; restoring one wire does not automatically restore every required connection.

T11

Partial credit allowed: True. Verify a simple conducting route and the encountered contact order under the specified states. Accept a different valid route. A valid unambiguously incomplete prefix may receive partial credit only if no invalid ordering or nonexistent step is asserted; an impossible complete path is incorrect.

T12

Partial credit allowed: False. Determine series/parallel/separate relative to the question endpoints and state scope. Use the task reference constraints; do not infer the relationship from device names alone.

T13

Partial credit allowed: False. Decide whether the given observation uniquely identifies the stated fault. One compatible fault is not proof of uniqueness. Preserve intact-wire and single-fault assumptions.

T14

Partial credit allowed: False. Assess whether the available schematic and explicit conditions establish the requested electrical voltage/state. A nameplate such as DC220V is not itself proof of live potential. Correct insufficiency is a valid answer, not an abstention.

T15

Partial credit allowed: False. Determine whether the specified contact opening interrupts the asked branch/route under the exact conditions. Do not confuse interrupting one route with disconnecting every possible parallel path.

S9. Evidence criteria and interpretation

Evidence is assessed for named-entity grounding, path or blocking-fact validity, compliance with explicit states, necessary alternatives and support for the answer. Equivalent valid routes or minimum plans are accepted. A correct answer with incorrect supporting details may receive answer credit but cannot receive joint credit. These scores measure observable reference-supported explanation quality, not the faithfulness of hidden computation.

S10. Reproducibility and release scope

Data and code: https://github.com/Talulah81920/ELEC-VQA. Dataset license: CC BY 4.0; original code: MIT. Original raw ratings and the separate score-completion records are preserved. The current manuscript, final tables, source files and file hashes accompany this submission package. Training uses one seed. Test questions share source drawings, and controlled variants are not additional independent original circuit designs. Reference graphs can be self-consistent despite extraction errors; native schematic edits do not establish field-fault prevalence or safe maintenance procedures.